

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC404 - WATER RESOURCES ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
1. What is the primary purpose of irrigation?
 - a) Increase rainfall
 - b) Supply water to crops
 - c) Drain excess water
 - d) Improve soil texture
 2. Duty of water is defined as:
 - a) Quantity of water required per hectare
 - b) Area irrigated by one unit of water in a given time
 - c) Depth of water applied to the crop
 - d) Time taken to irrigate a field
 3. The mass curve is used to determine:
 - a) Reservoir storage capacity
 - b) Dam height
 - c) Flow velocity
 - d) Sediment deposition
 4. Silting in reservoirs primarily affects:
 - a) Water quality
 - b) Storage capacity
 - c) Evaporation rate
 - d) Dam safety
 5. Useful life of a reservoir depends mainly on:
 - a) Sedimentation rate
 - b) Dam height
 - c) Water demand
 - d) Evaporation losses

- 6 Which dam type is designed to transfer water pressure horizontally to its abutments?
 - a) Gravity dam
 - b) Earth dam
 - c) Arch dam
 - d) Barrage
- 7 A spillway is primarily used for:
 - a) Water storage
 - b) Flood control by safely releasing excess water
 - c) Irrigation
 - d) Power generation
- 8 Kennedy's theory is related to:
 - a) Silt transportation in canals
 - b) Stability of dams
 - c) Design of weirs
 - d) Rainfall measurement
- 9 Canal lining is necessary to:
 - a) Increase seepage losses
 - b) Reduce seepage losses and increase efficiency
 - c) Increase evaporation
 - d) Strengthen the dam
- 10 Which of the following is NOT part of the hydrologic cycle?
 - a) Evaporation
 - b) Precipitation
 - c) Photosynthesis
 - d) Infiltration
- 11 Unit hydrograph is used to:
 - a) Estimate rainfall intensity
 - b) Determine runoff response to rainfall
 - c) Measure evaporation
 - d) Design dams
- 12 Lift irrigation differs from canal irrigation because:
 - a) Water is lifted using mechanical means
 - b) It uses natural gravity flow
 - c) It depends on rainfall only
 - d) It has no pumps

Q2. Solve the following.

- A) Derive relationship between duty and delta. 6
- B) The Gross Command Area (GCA) of an irrigation canal is 1,20,000 ha. The Culturable Command Area (CCA) is 75% of GCA and intensities of irrigation of kharif and rabi crops are 40 and 55 % respectively. If the duties of water at the canal are 800 and 1550 ha for kharif and rabi crops respectively. Determine the head discharge of the canal.

Q3. Solve the following.

- A) Explain the procedure of computation of reservoir capacity. 6
- B) Discuss in detail (i) Mass curve (ii) Demand curve 6

Q4. Solve Any Two of the following.

- A) Define the elementary profile of a gravity dam and discuss its importance in design. 6
- B) Explain Bligh's creep theory and state its limitations. 6
- C) Determine the section of Lacey's regime channel to carry $15 \text{ m}^3/\text{s}$ of discharge in an alluvial soil with a silt factor of 1.0. Use Lacey's regime equations to compute the regime velocity, area of cross-section, wetted perimeter, and bed slope of the channel. 6
- Q5. Solve Any Two of the following.
- A) What is precipitation? Describe various forms of precipitation. 6
- B) List the different methods used to compute the average depth of precipitation over a catchment area. Describe the procedure of any one method in detail. 6
- C) In a catchment area, rainfall data is available from five existing rain gauge stations. The average annual rainfall values (in mm) recorded at these stations are 850, 720, 600, 550, and 470 mm. If the permissible percentage error in the estimation of mean rainfall is 12%, determine the optimum number of rain gauges required in the catchment area. 6
- Q6. Solve Any Two of the following.
- A) What is water logging?. Suggest suitable preventive and remedial measures to control its adverse effects. 6
- B) Discuss in detail the concept of lift irrigation. 6
- C) What is rainwater harvesting? Explain its working principle with a suitable sketch. 6

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